



LED-BASED RUNNING DISPLAY

■ N. PUGALUM PERUMAL

Here is a circuit that creates an eye-catching running display effect using LEDs. It can be used to light up borders of animations, pictures, etc. and also for short word displays.

The circuit is built around an NE555 timer (IC1), a decade counter IC CD4017 (IC2) and a few discrete

components. IC1 is wired as an astable multivibrator whose output is fed to the clock input (pin 14) of the counter.

When power to the circuit is switched on, NE555 oscillates to produce clock pulses. These pulses are fed to the clock input (pin 14) of the counter, which starts counting from Q0 to Q5. When output Q0 of the counter goes high, line L1 (LED1 through LED6) glows. When output Q1 of the counter goes high,

lines L1 and L2 (LED7 through LED12) glow sequentially. When output Q2 goes high, lines L1, L2 and L3 (LED13 through LED18) glow sequentially. When output Q3 goes high, L1, L2, L3 and L4 (LED19 through LED24) glow sequentially.

When outputs Q4 and Q5 of the counter go high, L1, L2, L3 and L4 (LED1 through LED24) remain lit-up. Further, when output Q6 goes high, the counter automatically resets and the process repeats.

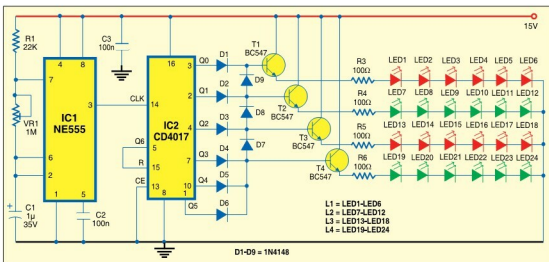
The LED status based on the counter output is shown in the table. Using this table, you can display any word with a running effect. Suppose you want to display 'STAR.' Build letter S with L1, T with L2, A with L3 and R with L4. When the circuit is switched on, letter S glows first. Then S and T glow sequentially, followed by S, T and A, and then S, T, A and R. When Q4 and Q5 outputs

of the counter go high, the complete word 'STAR' glows. Thereafter, letter 'S' glows again and the cycle repeats.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Arrange the group of five LEDs as desired—for word display or the border of a pictorial animation. ●

LED Status Based on the Counter Output

Output of CD4017	L1 (LED1-LED6)	L2 (LED7-LED12)	L3 (LED13-LED18)	L4 (LED19-LED24)
Q0 = High	On	Off	Off	Off
Q1 = High	On	On	Off	Off
Q2 = High	On	On	On	Off
Q3 = High	On	On	On	On
Q4 = High	On	On	On	On
Q5 = High	On	On	On	On
Q6 = High	On	Off	Off	Off



DOOR-OPEN ALARM

■ PRADEEP G.

This door-open alarm uses an input sensor fixed at the corner of the door frame and a magnet

fixed on the door panel close to the sensor. When the door is opened, the magnet moves away from the sensor, generating a controlling signal to sound the alarm.



The circuit is built around Hall sensor A1301 (IC1), timer NE555